Merge script

2025-03-24

```
res_dir <- "../results/s01_qc_merge/"
```

This script merges our data with additional data from Madagascar, Borneo, Sumba, and 1000 Genomes Project, creating two datasets: one Malagasy only and another including Malagasy alongside all additional populations.

Build of each dataset

Madagascar_full: GRC37 (hg19) 1000_Genomes_b37_clean: GRC37 (hg19) NEO_full: GRC37 (hg19) - build noted in metadata Pierron_Malagasy: GRC37 (hg19) - verified using `bigsnpr::snp_modifyBuild()` Borneo_raw_data: GRC37 (hg19) - verified using `bigsnpr::snp_modifyBuild()` GSE76645_INDONESIA_SUMBA_TIMO GRC37 (hg19) - verified using `bigsnpr::snp_modifyBuild()`

Note: `snp_modifyBuild` calls LiftOver from UCSC (downloaded at https://hgdownload.cse.ucsc.edu/admin/exe/linux.x86_64/liftOver) to perform the build conversion. For datasets where build is not verified, this function is used to check chromosome, position, and rsID against data with known build.

Generate gen tibble

```
if (file.exists("../data/raw/data_input/Madagascar_full.gt")) {
  madagascar <- gt_load("../data/raw/data_input/Madagascar_full.gt")
} else {
  madagascar <- gen_tibble("../data/raw/data_input/Madagascar_full.bed",
    quiet = TRUE,
    valid_alleles = c("T", "G", "A", "C", "I"),
    allow_duplicates = TRUE
  )
}

cat("Number of individuals:", nrow(madagascar), "\n")
```

```
## Number of individuals: 192
```

```
cat("Number of loci:", count_loci(madagascar))
```

```
## Number of loci: 713014
```

```
if (file.exists("../data/raw/data_input/Madagascar_full_v2.bk")) {
  file.remove("../data/raw/data_input/Madagascar_full_v2.gt")
  file.remove("../data/raw/data_input/Madagascar_full_v2.bk")
  file.remove("../data/raw/data_input/Madagascar_full_v2.rds")
}
```

Our raw dataset has 713014 SNPs and 192 individuals

Add metadata

```
# Read in Pierron Malagasy metadata
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

# Add "RR" to pop_table$IID
pop_table$IID <- paste0(pop_table$IID, "RR")

# Find Malagasy entries
pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar$id & pop_table$V5 == "Madagascar"
  )

# match up the ID's with pop table metadata
matching_m <- match(madagascar$id, pop_table_sub$IID)

madagascar$population <-
  ifelse(!is.na(matching_m),
    pop_table_sub$Population[matching_m], madagascar$population
  )
```

Remove the “I” loci

```
remove <- which(show_loci(madagascar)$allele_ref == "I")
madagascar <- madagascar %>% select_loci(-all_of(c(remove)))
count_loci(madagascar)
```

```
## [1] 713013
```

Subset to autosomes only

```
madagascar <-
  madagascar %>% select_loci_if(show_loci(genotypes)$chromosome %in% c(1:22))
count_loci(madagascar)
```

```
## [1] 693051
```

Find and remove duplicates

```
dups <- find_duplicated_loci(madagascar)
madagascar <- madagascar %>% select_loci(-all_of(dups))
count_loci(madagascar)
```

```
## [1] 693045
```

```
# remove entries with genetic distance == 0
no_dist <- which(show_loci(madagascar)$genetic_dist == 0)
madagascar <- madagascar %>% select_loci(-all_of(no_dist))
count_loci(madagascar)
```

```
## [1] 691064
```

Malagasy Pierron data

```
if (file.exists("../data/raw/data_input/Pierron_Malagasy.gt")) {
  madagascar_p <- gt_load("../data/raw/data_input/Pierron_Malagasy.gt")
} else {
  madagascar_p <-
    gen_tibble("../data/raw/data_input/Pierron_Malagasy.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

cat("Number of individuals:", nrow(madagascar_p), "\n")
```

```
## Number of individuals: 69
```

```
cat("Number of loci:", count_loci(madagascar_p))
```

```
## Number of loci: 661324
```

Append population info

```
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar_p$id & pop_table$V5 == "Madagascar"
  )

matching_m <- match(madagascar_p$id, pop_table_sub$IID)
madagascar_p$population <- ifelse(!is.na(matching_m),
  pop_table_sub$Population[matching_m],
  madagascar_p$population
)
```

Find and remove duplicates

```
dups <- find_duplicated_loci(madagascar_p)
madagascar_p <- madagascar_p %>% select_loci(-all_of(dups))

# remove entries with no allele_ref & no allele_alt
no_allele <- which(is.na(show_loci(madagascar_p)$allele_ref) & is.na(show_loci(madagascar_p)$allele_alt
madagascar_p <- madagascar_p %>% select_loci(-all_of(no_allele))
count_loci(madagascar_p)
```

```
## [1] 658101
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge.rds")
}
```

```
## [1] TRUE
```

Merge the two datasets

```
malagasy_merged <- rbind(madagascar, madagascar_p,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/malagasy_merge"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
```

```
## ----------------------------
## dataset: reference
## number of SNPs: 691064 reduced to 642520
## ( 1954 are ambiguous, of which 1954  were removed)
## ----------------------------
## dataset: target
## number of SNPs: 658101 reduced to 642520
## ( 171 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mala

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_v2.bk")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_v2.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_v2.rds")
}
```

```
## [1] TRUE
```

Sort the order

```
# reorder the loci
malagasy_merged <- gt_order_loci(malagasy_merged,
  use_current_table = FALSE,
  ignore_genetic_dist = TRUE
)
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

Quality control on the merged dataset

First we create a gt 'malagasy_merge_clean_no_maf_1st_degree' for ROH analysis, this data will include all but first degree relatives and is not MAF filtered.

```
malagasy_merged <- malagasy_merged %>% group_by(population)
loci_report_m <- qc_report_loci(malagasy_merged)
malagasy_merged <- malagasy_merged %>% ungroup()
```

Check HWE Merina vs other groups

```
malagasy_merged <- malagasy_merged %>% group_by(population)
hwe_res <- loci_hwe(malagasy_merged)

hwe_dist_per_pop <- hwe_res %>% pivot_wider(names_from = group, values_from = value)
merina <- as.data.frame(hwe_dist_per_pop$Merina)
colnames(merina) <- "Value"

merina_hwe <- ggplot(merina, aes(x = Value)) +
  geom_density() +
  labs(title = "Merina HWE")

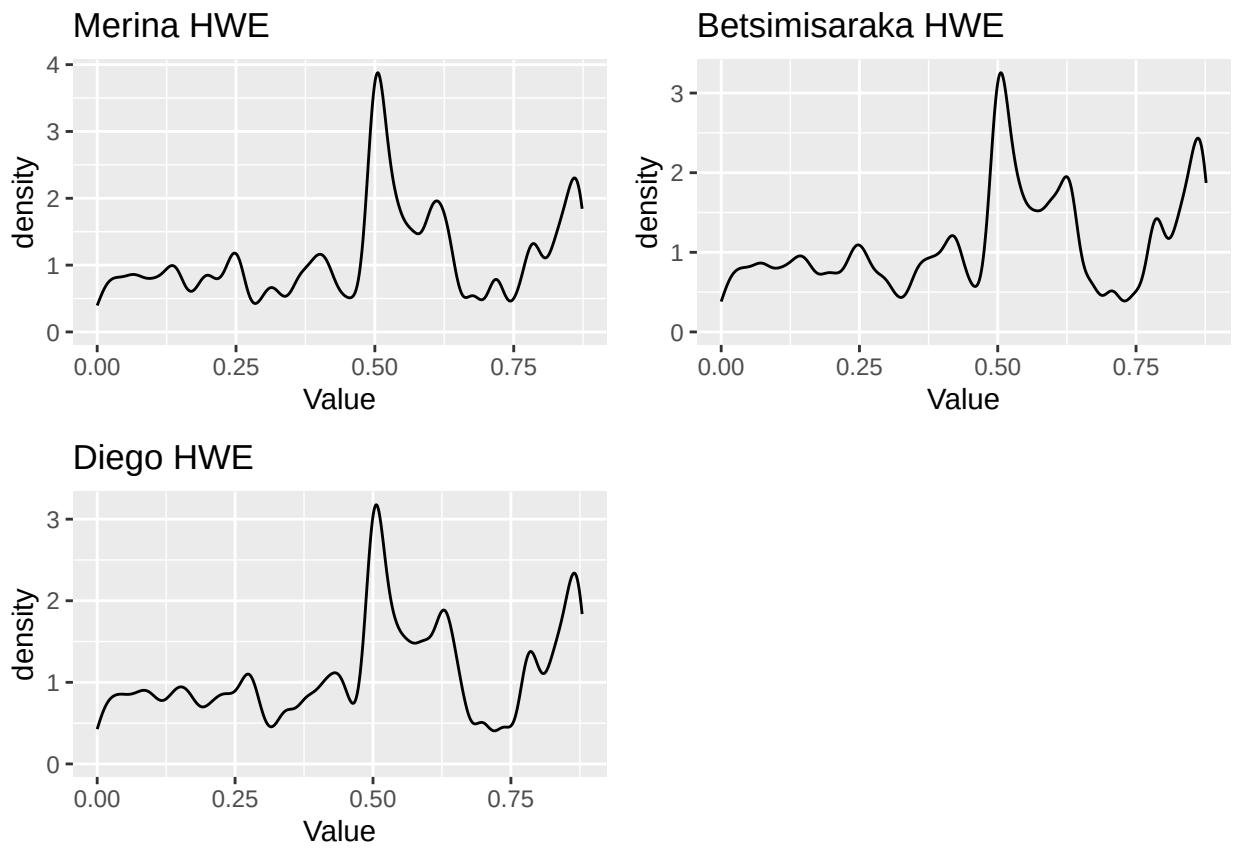
betsimisaraka <- as.data.frame(hwe_dist_per_pop$Betsimisaraka)
colnames(betsimisaraka) <- "Value"

betsimisaraka_hwe <- ggplot(betsimisaraka, aes(x = Value)) +
  geom_density() +
  labs(title = "Betsimisaraka HWE")

diego <- as.data.frame(hwe_dist_per_pop$Diego)
colnames(diego) <- "Value"

diego_hwe <- ggplot(diego, aes(x = Value)) +
  geom_density() +
  labs(title = "Diego HWE")

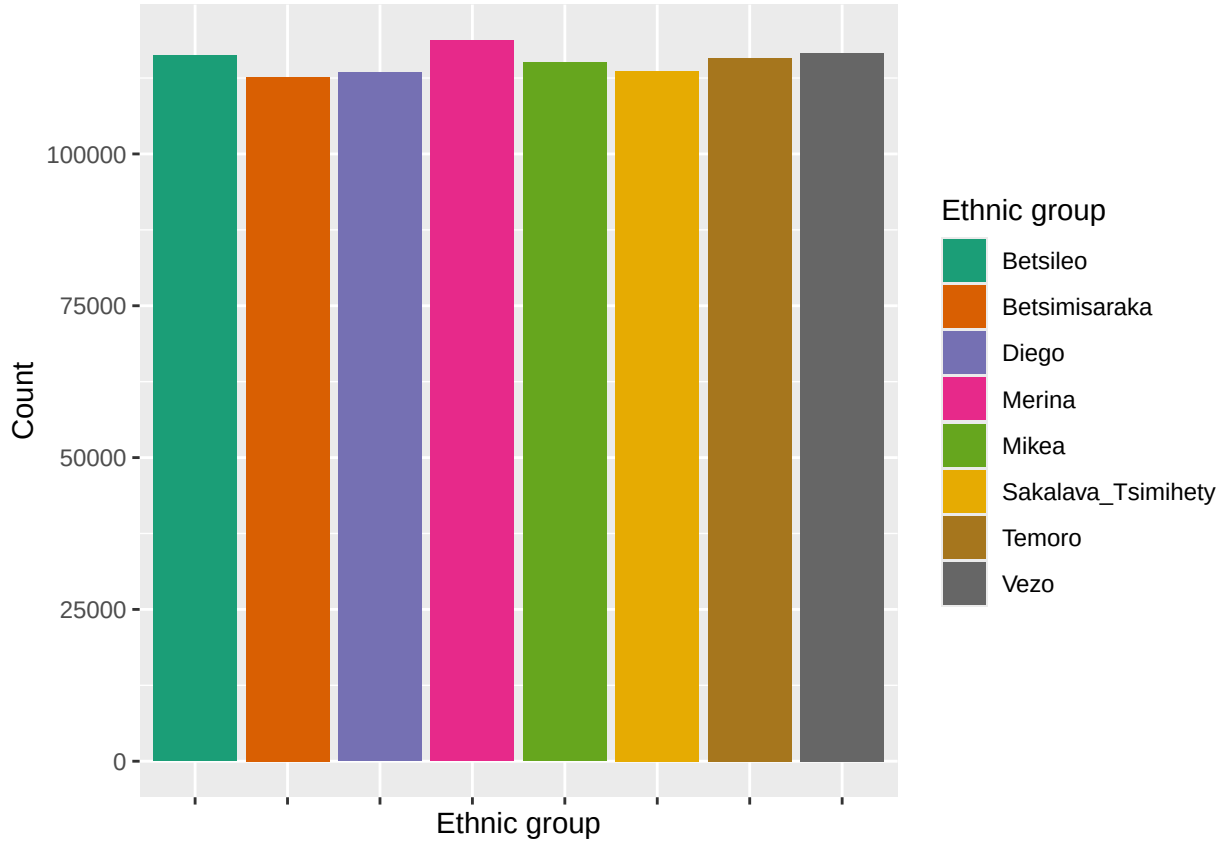
gridExtra::grid.arrange(merina_hwe, betsimisaraka_hwe, diego_hwe, ncol = 2, nrow = 2)
```



Shape of distribution of HWE p-values looks similar across groups.

```
min_vals <- hwe_res %>%
  mutate(value = value * length(unique(group))) %>% # correction
  group_by(loci) %>%
  slice_min(value, n = 1)

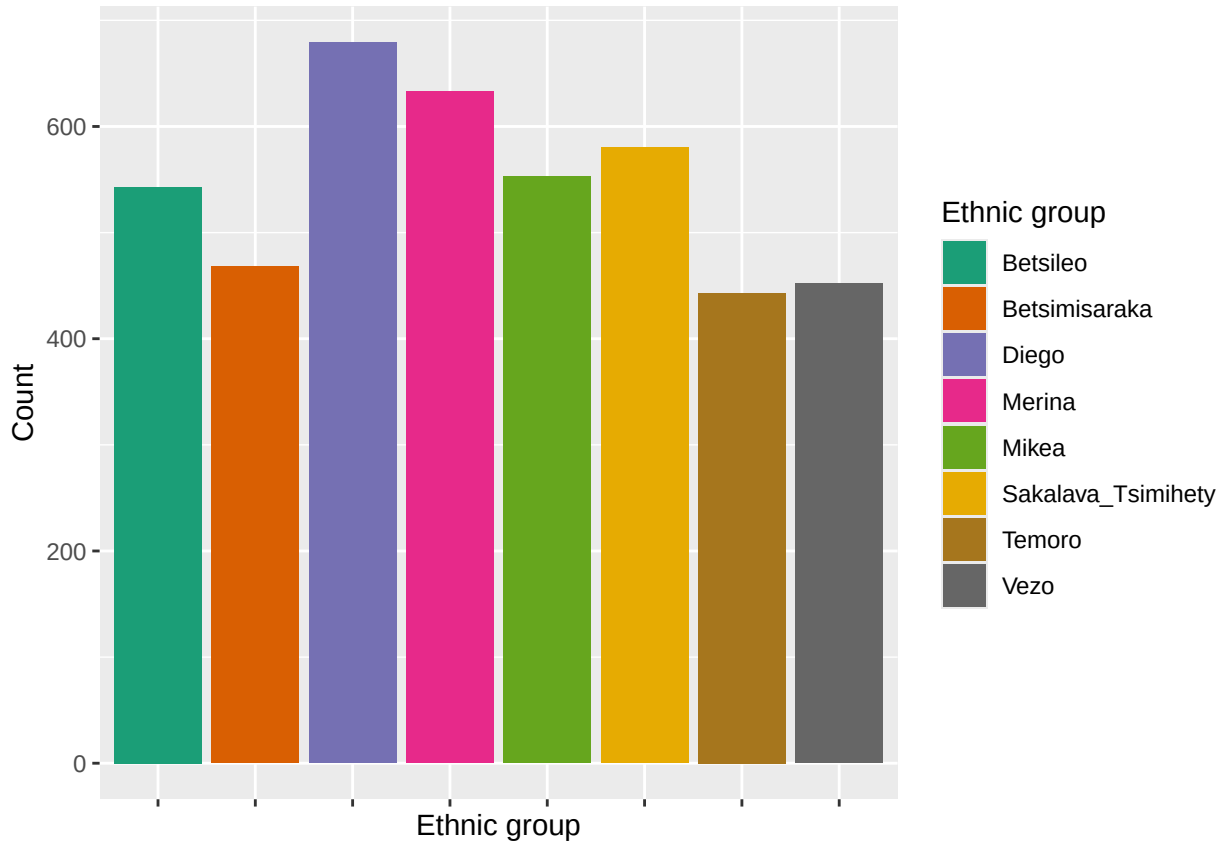
ggplot(min_vals, aes(x = group, fill = group)) +
  geom_bar() +
  theme(axis.text.x = element_blank()) +
  labs(x = "Ethnic group", fill = "Ethnic group", y = "Count") +
  scale_fill_brewer(palette = "Dark2")
```



Minimum p-values are distributed evenly across groups.

```
min_vals <- hwe_res %>%
  mutate(value = value * length(unique(group))) %>% # correction
  filter(value < 0.01)

ggplot(min_vals, aes(x = group, fill = group)) +
  geom_bar() +
  theme(axis.text.x = element_blank()) +
  labs(x = "Ethnic group", fill = "Ethnic group", y = "Count") +
  scale_fill_brewer(palette = "Dark2")
```



P-values < 0.01 are not elevated in the Merina.

Filter loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
malagasy_merged <- malagasy_merged %>% select_loci(all_of(to_keep))
count_loci(malagasy_merged)
```

```
## [1] 638245
```

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(malagasy_merged)
```

```
## [1] 620093
```

Filter individuals

```
malagasy_merged <- malagasy_merged %>% group_by(population)
individual_report_m <- qc_report_indiv(malagasy_merged, kings_threshold = 0.177)
malagasy_merged <- malagasy_merged %>% ungroup()
malagasy_merged <-
  malagasy_merged %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
malagasy_merged <-
  malagasy_merged %>% filter(indiv_missingness(genotypes) < 0.1)
```

Loci and sample count

```
nrow(malagasy_merged)
```

```
## [1] 243
```

```
count_loci(malagasy_merged)
```

```
## [1] 620093
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.rds")
}
```

```
## [1] TRUE
```

```
#gt_save(malagasy_merged,
#  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree"
#)
gt_update_backingfile(malagasy_merged,
  backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree"
)
```

```
##
## gen_backing files updated, now
```

```
## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy
```

```
## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m
```

```
## make sure that you do NOT delete those files!
```

```
## to reload the gen_tibble in another session, use:
```

```
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

```
## # A gen_tibble: 620093 loci
## # A tibble:     243 x 3
##    id       population     genotypes
##    <chr>    <chr>         <vctr_SNP>
##  1 BET001RR Betsimisaraka  [0,0,...]
##  2 BET002RR Betsimisaraka  [1,0,...]
##  3 BET003RR Betsimisaraka  [0,0,...]
##  4 BET004RR Betsimisaraka  [0,0,...]
##  5 BET005RR Betsimisaraka  [0,0,...]
##  6 BET006RR Betsimisaraka  [1,0,...]
##  7 BET007RR Betsimisaraka  [0,0,...]
##  8 BET009RR Betsimisaraka  [0,0,...]
##  9 BET010RR Betsimisaraka  [0,1,...]
## 10 BET011RR Betsimisaraka  [0,0,...]
## # i 233 more rows
```

Next we add MAF filtering and remove second degree relatives to create gt 'malagasy_merge_clean' for other analysis.

Add MAF filtering

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(malagasy_merged)
```

```
## [1] 579828
```

Add second degree relatedness filtering

```
malagasy_merged <- malagasy_merged %>% group_by(population)
individual_report_m <-
  qc_report_indiv(malagasy_merged, kings_threshold = 0.0884)
malagasy_merged <- malagasy_merged %>% ungroup()
malagasy_merged <-
  malagasy_merged %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
nrow(malagasy_merged)
```

```
## [1] 233
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.rds")
}
```

```
## [1] TRUE
```

```
#gt_save(malagasy_merged,
#  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean"
#)
malagasy_merged <-
  gt_update_backingfile(malagasy_merged,
    backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

Impute

```
malagasy_merged <- gt_impute_simple(malagasy_merged, method = "mean")
```

LD clumping

```
to_keep_LD <- loci_ld_clump(malagasy_merged,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)
malagasy_merged_ld <- malagasy_merged %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(malagasy_merged_ld)
```

```
## [1] 233
```

```
count_loci(malagasy_merged_ld)
```

```
## [1] 224001
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.rds")
}
```

```
## [1] TRUE
```

Save imputed LD filtered dataset

```
#gt_save(malagasy_merged_ld,
#  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean_ld"
#)
malagasy_merged <-
  gt_update_backingfile(malagasy_merged_ld,
    backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean_ld"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

Load Borneo data

```
if (file.exists("../data/raw/data_input/Borneo_raw_data.gt")) {
  borneo <- gt_load("../data/raw/data_input/Borneo_raw_data.gt")
} else {
  borneo <-
    gen_tibble("../data/raw/data_input/Borneo_raw_data.bed",
      valid_alleles = c("A", "T", "C", "G", "I", "N"),
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

borneo$population <- "Borneo"
```

Find and remove duplicates

```
dups <- find_duplicated_loci(borneo)
borneo <- borneo %>% select_loci(-all_of(dups))
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_borneo.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.rds")
}
```

```
## [1] TRUE
```

Merge

```
malagasy_merged <-
  gt_load("../data/intermediate/rbind_data/malagasy_merge_clean.gt")
malagasy_borneo <- rbind(malagasy_merged, borneo,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/malagasy_borneo"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 579828 reduced to 510925
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 715316 reduced to 510925
## ( 93589 were flipped to match the reference set)
## ( 1798 are ambiguous, of which 1798 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mala

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_borne
```

Load 1000 Genomes data

Cutting down to populations of interest

```
if (file.exists("../data/raw/data_input/1000_Genomes_b37_clean.gt")) {
  g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
} else {
  g1000 <-
    gen_tibble("../data/raw/data_input/1000_Genomes_b37_clean.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

# append population info
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

pop <- c("KHV", "BEB", "YRI", "PJL", "GWD", "MSL", "ESN", "LWK", "CHS")
#- Kinh in Ho Chi Minh City, Vietnam
#- Bengali in Bangladesh
#- Han Chinese South
#- Luhya in Webuye, Kenya
#- Yoruba in Ibadan, Nigeria
#- Punjabi in Lahore, Pakistan
#- Gambian in Western Division, The Gambia - Mandinka
#- Mende in Sierra Leone
#- Esan in Nigeria

g1000 <- g1000 %>% filter(population %in% pop)

count_loci(g1000)
```

```
## [1] 670552
```

```
nrow(g1000)
```

```
## [1] 882
```

Find and remove duplicates

```
dups <- find_duplicated_loci(g1000)
g1000 <- g1000 %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000 <- rbind(malagasy_borneo, g1000,
  as_is = FALSE, flip_strand = TRUE, use_position = FALSE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/mad_bor_g1000"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 510925 reduced to 502662
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 670552 reduced to 502662
## ( 91215 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000.
```

Load Sumba data

```r
if (file.exists("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")) {
  sumba <- gt_load("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")
} else {
  sumba <-
    gen_tibble("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}

sumba$population <- "Sumba"

timor <- read.table("../data/raw/data_input/timor")
timor$population <- "Timor"

matching_m <- match(sumba$id, timor$V2)
sumba$population <- ifelse(!is.na(matching_m),
  timor$population[matching_m],
  sumba$population
)

count_loci(sumba)
```

```
## [1] 713246
```

```r
nrow(sumba)
```

```
## [1] 240
```

```r
set.seed(1)

subsample <- function(x) {
  if (nrow(x) <= 30) {
    return(x)
  } else {
    sample_n(x, size = 30)
  }
}

sumba_only <- sumba %>%
  group_by(population) %>%
  group_modify(~ subsample(.x)) %>%
  ungroup()

sumba <- sumba %>% filter(id %in% sumba_only$id)

nrow(sumba)
```

```
## [1] 37
```

Find and remove duplicates

```
dups <- find_duplicated_loci(sumba)
sumba <- sumba %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000_sum <- rbind(mad_bor_g1000, sumba,
  as_is = FALSE,
  flip_strand = TRUE,
  quiet = FALSE,
  use_position = FALSE,
  backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 502662 reduced to 502646
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 713246 reduced to 502646
## ( 91213 were flipped to match the reference set)
## ( 2056 are ambiguous, of which 2056 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_
```

Load NEO data

```
if (file.exists("../data/intermediate/rbind_data/NEO_full.gt")) {
  NEO_full <- gt_load("../data/intermediate/rbind_data/NEO_full.gt")
} else {
  NEO_full <-
    gen_tibble("../data/raw/data_input/NEO_full.bed",
      quiet = TRUE,
      allow_duplicates = TRUE
    )
}
```

Find and remove duplicates

```
dups <- find_duplicated_loci(NEO_full)
NEO_full <- NEO_full %>% select_loci(-all_of(dups))
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000_sum_neo <- rbind(mad_bor_g1000_sum, NEO_full,
  as_is = FALSE,
  flip_strand = TRUE,
  quiet = FALSE,
  use_position = TRUE,
  backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ----------------------------
## dataset: reference
## number of SNPs: 502646 reduced to 418687
## ( 0 are ambiguous, of which 0  were removed)
## ----------------------------
## dataset: target
## number of SNPs: 1400891 reduced to 418687
## ( 1 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)
```

```
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_
```

Quality control on the merged dataset

```
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
loci_report_m <- qc_report_loci(mad_bor_g1000_sum_neo)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
```

Filter loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% select_loci(all_of(to_keep))
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 417980
```

```
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 237449
```

```
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 235750
```

Filter indiv

```r
# temporarily remove all populations with only one individual
pop_tbl <- table(mad_bor_g1000_sum_neo$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove) # this removes a lot of DRC samples and some Mozambique samples (rel

removed_pops <- mad_bor_g1000_sum_neo %>%
  filter(population %in% to_remove_pops)

mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(!population %in% to_remove_pops)


# filter indivs
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
individual_report_m <-
  qc_report_indiv(mad_bor_g1000_sum_neo, kings_threshold = 0.0884)
mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)

if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_rebind.bk")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_rebind.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_rebind.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_rebind.rds")
}
```

```
## [1] TRUE
```

```r
# bind them together again
mad_bor_g1000_sum_neo <- rbind(mad_bor_g1000_sum_neo, removed_pops, backingfile = "../data/intermediate/
```

```
## harmonising loci between two datasets
## flip_strand =  FALSE  ; remove_ambiguous =  FALSE
## ------------------------------
## dataset: reference
## number of SNPs: 235750 reduced to 235750
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 235750 reduced to 235750
## ( 0 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!
```

```
## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_s
```

```
NEO_full <-
  NEO_full %>% filter(indiv_missingness(genotypes) < 0.1)

# remove all populations with only one individual
# pop_tbl <- table(mad_bor_g1000_sum_neo$population)
# to_remove <- which(pop_tbl == 1)
# to_remove_pops <- names(to_remove) # this removes a lot of DRC samples and some Mozambique samples (r
# mad_bor_g1000_sum_neo <-
#   mad_bor_g1000_sum_neo %>% filter(!population %in% to_remove_pops)

# run filtering
# mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% group_by(population)
# individual_report_m <-
#   qc_report_indiv(mad_bor_g1000_sum_neo, kings_threshold = 0.0884)
# mad_bor_g1000_sum_neo <- mad_bor_g1000_sum_neo %>% ungroup()
# mad_bor_g1000_sum_neo <-
#   mad_bor_g1000_sum_neo %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TR
mad_bor_g1000_sum_neo <-
  mad_bor_g1000_sum_neo %>% filter(indiv_missingness(genotypes) < 0.1)
```

Loci and sample count

```
nrow(mad_bor_g1000_sum_neo)
```

```
## [1] 2742
```

```
count_loci(mad_bor_g1000_sum_neo)
```

```
## [1] 235750
```

Save

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean.rds")
}
```

```
## [1] TRUE
```

```
#gt_save(
#  mad_bor_g1000_sum_neo,
#  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean"
#)
mad_bor_g1000_sum_neo <-
  gt_update_backingfile(mad_bor_g1000_sum_neo,
    backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean"
  )
```

```
##
## gen_backing files updated, now

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_s
```

Impute

```
mad_bor_g1000_sum_neo <- gt_impute_simple(mad_bor_g1000_sum_neo,
  method = "mean"
)
```

LD clumping

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.rds")
}
```

```
## [1] TRUE
```

```
to_keep_LD <- loci_ld_clump(mad_bor_g1000_sum_neo,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)

mad_bor_g1000_sum_neo_ld <- mad_bor_g1000_sum_neo %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(mad_bor_g1000_sum_neo_ld)
```

```
## [1] 2742
```

```
count_loci(mad_bor_g1000_sum_neo_ld)
```

```
## [1] 138753
```

Save LD filtered dataset

```
#gt_save(
#  mad_bor_g1000_sum_neo_ld,
#  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
#)
mad_bor_g1000_sum_neo_ld <-
  gt_update_backingfile(mad_bor_g1000_sum_neo_ld,
    backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
  )
```

```
##
## gen_backing files updated, now
```

```
## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g
```

```
## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m
```

```
## make sure that you do NOT delete those files!
```

```
## to reload the gen_tibble in another session, use:
```

```
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_s
```

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.

Methods

Genetic and phenotypic data collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands, Betsileo people from the southern Highlands, Sakalava and Tsimihety people from the northwestern coast, Betsimisaraka people from the eastern coast, and people of various ethnic groups from the extreme north in the town of Diego. Samples were collected from married couples, and ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global dataset combining Malagasy samples with comparative data. The Malagasy dataset was created by merging the data from this study with Malagasy data from Pierron et al. (2014) (n = 69). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the 1000 Genomes Project Consortium (2015) dataset (n = 882), individuals from Borneo from Brucato et al. (2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from 37) and Timor (n = 7) from Cox et al. (2016).

Data were cleaned and merged in R with package tidypopgen (Carter and Manica, 2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.01 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. All SNPs with minor allele frequency below a threshold of 0.025 were removed. All SNPs with a Hardy-Weinberg equilibrium test p-value below 0.01 were removed. Close relatives were removed using the KING robust kinship coefficient (Manichaikul et al., 2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 579828 SNPs, and the global dataset consisted of 2742 individuals and 235750 SNPs.

SNPs in linkage were then removed by LD clumping (Privé et al., 2018), using tidypopgen. Those with an r2 value greater than 0.2 in a sliding window of 100 kilobases were removed. After pruning, the Malagasy dataset consisted of 224001 SNPs and the global dataset consisted of 138753 SNPs.